CHARUSAT UNIVERSITY OF SCIENCE AND TECHNOLOGY
M. Sc. BIOCHEMISTRY (SEMESTER II)
EXTERNAL EXAMINATION
BC 705 - IMMUNOLOGY

DATE: 28.04.2010
TIME: 10.00AM TO 1.00AM **TOTAL MARKS:20**

REGISTRATION NUMBER OF THE CANDIDATE :

MARKS OBTAINED IN PART A: **SIGNATURE OF THE SUPERVISOR**

TIME DURATION: 30MINS (10.00-10.30)

(PART A)

1. γδ TCR recognize Ag in MHC independent manner. **(True / False)**

2. A precipitate of antigen and antibody can be formed with monovalent Fab fragment. **(True / False)**

3. TAP is required for antigen presentation in both MHC I and MHC II pathways. **(True / False)**

4. Immunoelectrophoresis is a quantitative method. **(True / False)**

5. CD3 complex is associated with TCR for activation of Tcell. **(True / False)**

6. The immunoglobulin class that binds with very high affinity to membrane receptors on basophils and mast cells is:
   a)IgG b) IgA c) IgD d) IgE

7. The mechanism that permits different immunoglobulins isotypes to be synthesized is:
   a) allelic exclusion b)codominant expression c)class switching d) differential RNA processing

8. The valency of IgM secreted by plasma B cell is:
   a) 2 b)4 c)8 d) 10

9. In all the complement pathways C5 convertase join with C 6,7,8,9 to form
   a) membrane attack complex, b) Complement activation complex, c) Initiation complex, d) None of the above.

10. Surrogate light chain is seen in ------- stage B cell development:
    a) Pro B b) Pre-B c) Immature B cell d) None of the above

11. Which of the following statement is not true for MHC?
    a) MHC I and II molecules are membrane-bound
    b) Mature T cell must have TCR that recognizes particular MHC
    c) Cytokines (especially IFN-γ) increase expression of MHC
    d) MHC genes follow allelic exclusion

12. ------ TCR are enriched at mucosal surfaces
   a) αβ, b) γε, c) βδ, d) γδ

13. The major role of T cells in the immune response includes which one of the following?
   a) Recognition of epitopes presented with major histocompatibility complex molecules on all surfaces
   b) Complement fixation c)Phagocytosis d)Production of antibodies

14. Which one of the following anatomical regions is most likely to show a predominance of T lymphocytes?
   a) A periarteriolar sheet in the spleen b) A Peyer's patch in the small intestine c)A tonsilar follicle d) The bone marrow

15. secondary lymphoid organs are required for:
   a) maturation of B and T lymphocytes b)exposing the antigens to immature B and T lymphocytes c)both a and b d)none of the above

16. Immunoglobulin gene segment rearrangement follows:
   a) one-turn / two-turn joining rule b) two-turn / two-turn joining rule c) one-turn / one-turn joining rule d) none of the above

17. The antigen binding sites of an antibody molecule are determined by the structure of the: a) Constant region of heavy chains b)Constant region of light chains c)Variable region of heavy chains d) Variable regions of both heavy and light chains

18. Self reacting B cells sustain bone marrow selection by
   a) class switching, b) affinity maturation, c)light chain editing, d) all of the above.

19. During B cell activation ITAM is required for
   a) signal transduction, b)ligand binding, c)activate tyrosine kinases, d)none of the above

20. Which of the following does not participate in the formation of antigen-antibody complexes?
   a) Hydrophobic bonds b) Covalent bonds c) Electrostatic interactions d) Hydrogen bonds

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(PART B)

Section I

Q.1(A) Describe the transportation of IgA across the epithelial membrane. 3

Q.1(B) describe the mechanism of immunoglobulin gene recombination by Rag 1 and 2 using one and two turn sequences. 5

Q.1(C) Explain the terms Allelic exclusion and Class switching. 2

OR

Explain **any two** of the following allotype, isotype and idiotype with respect to antibody structure. 2

Q.2 (A)i. State any two uses of monoclonal antibodies 2

ii. What are the different types of interactions present in antigen antibody complex? 2

Q.2 (B) What is FACS? Give its advantages. 2

Q.2 (C)Highlight the method and significance of direct and indirect coombs test. 4

OR

What is the difference between agglutination and precipitation? Describe hemeagglutination with its use. 4

Section II

Q.3.(A) What is oxygen dependant killing of microorganism in phagocytosis? 4

Q.3(B) "Some micro organisms (*Neisseria gonorrhoeae*) are called intracellular pathogens". Justify 3

Q.3(C) Differentiate between innate and acquired immunity. 3

OR

Briefly outline the physiological barriers of innate immunity. 3

Q.4 (A) Describe the class-I MHC pathway of antigen processing and presentation. 4

Q.4 (B) What do you mean by positive and negative selection of thymocytes? Explain. 2

OR

What is B cell co receptor? Discuss 2

Q.4 (C) Describe the steps involved in T cell mediated cell death. 4

Q.5 (A) Differentiate between active and passive immunity with appropriate example each. 4

Q.5 (B) How does DNA vaccine work? Describe 4

OR

Explain the process involved in the release of effector molecules in Type-I hypersensitivity. 4

Q.5 (C) Give different classes of chemokines based on the amino acid position. 2